Database Technologies Assignment 3: Functional Dependencies and Normalization

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Task 1

Consider the relation schema R(A, B, C, D, E, F) and the following three FDs:

FD1: {A} → {B,C} FD2: {C} → {A,D} FD3: {D,E} → {F}

Use the Armstrong rules to derive each of the following two FDs. In both cases, describe the derivation process step by step (i.e., which rule did you apply to which FDs).

a) {C} → {B} b) {A,E} → {F}

a)

Derivation process using armstrong rules:

1. FD2: From {C} -> {A,D}, we know {C} -> {A}, with the projection rule, decomposing FD2.
2. FD1: From {A} -> {B,C}, we know {A} -> {B}, with the projection rule, decomposing FD1.
3. Since {C} -> {A} and {A} -> {B}, we derive {C} -> {B}, with the transitivity rule.

b)

Derivation Process (using Armstrong rules):

1. FD3: From {D,E} -> {F}, we know {D,E} determines F, given functional dependency.
2. FD2: From {C} -> {A,D}, we know {A,E} -> {D,E}, with the union rule combining {A} and {E} to create the determinant.
3. Since {A,E} -> {D,E} (derived from FD2) and {D,E} -> {F} (FD3), we derive {A,E} -> {F}.

Complementary work 1b):

Here we clarify why {A, E} -> {D, E} can be derived from FD2: {C} -> {A, D}, we know that {C} determines {A, D}. Since {A, E} already includes {A}, we can take this and say: {A, E} determines {A} (which is trivial since {A} is part of determinant). So by combining {A, E} with the fact that {A, D} can be determined with FD2, we know: {A, E} -> {D, E}, because adding {E} does not change the set of determined attributes. This logic shows why {A, E} -> {D, E} holds based on FD2.

Task 2

For the aforementioned relation schema with its functional dependencies, compute the attribute closure X^+ for each of the following two sets of attributes.

a) X = { A } b) X = { C, E }

a)
Attribute closure process gives:

1. Start with X^+ = {A}.
2. FD1: Since {A} -> {B,C}, add {B,C} to X^+.
   X^+ = {A,B,C}.
3. FD2: Since {C} -> {A,D}, add {D} to X^+.
   X^+ = {A,B,C,D}.

b)
Attribute closure process gives:

1. Start with X^+ = {C,E}.
2. FD2: Since {C} -> {A,D}, add {A,D} to X^+.
   X^+ = {C,E,A,D}.
3. FD3: Since {D,E} -> {F}, add {F} to X^+.
   X^+ = {C,E,A,D,F}.

Complementary work 2b):
Here we make it clear why attribute B should be included in the closure:
Starting with X^+ = {C, E}, we applied FD2: {C} -> {A, D} to add {A, D}, and FD3: {D, E} -> {F} to add {F}. But we must also apply FD1: {A} -> {B, C}:
And since {A} is included in the closure after applying FD2, and {A} determines {B, C}, we can add {B} to the closure. So B is part of X^+, so the full closure is: X^+ = {A, B, C, D, E, F}.

Task 3

Consider the relation schema R(A, B, C, D, E, F) with the following FDs

FD1: {A,B} → {C,D,E,F}

FD2: {E} → {F}

FD3: {D} → {B}

a) Determine the candidate key(s) for R.

b) Note that R is not in BCNF. Which FD(s) violate the BCNF condition?

c) Decompose R into a set of BCNF relations, and describe the process step by step (don't forget to determine the FDs and the candidate key(s) for all of the relation schemas along the way).

a)

1. Combine attributes until X^+ = R.
2. {A,B}^+ = {A,B,C,D,E,F}.
3. So {A,B} is the candidate key.

Complementary work for 3a):
To find the candidate keys we compute the closure of attribute sets:
Compute {A, B}^+: We start with {A, B}. Using FD1: {A, B} -> {C, D, E, F}, add {C, D, E, F}. {A, B}^+ = {A, B, C, D, E, F} = R. So {A, B} is a candidate key.
Compute {A, D}^+: Start with {A, D}. Using FD3: {D} -> {B}, add {B}. Using FD1: {A, B} -> {C, D, E, F}, add {C, E, F}. {A, D}^+ = {A, B, C, D, E, F} = R.
So {A, D} is also a candidate key. So the candidate keys are both {A, B} and {A, D}.

b)

A relation violates BCNF if a non-trivial FD exists where the determinant is not a superkey.

1. FD2: {E} -> {F} (Determinant {E} is not a superkey).
2. FD3: {D} -> {B} (Determinant {D} is not a superkey).

c)

1. Decompose using FD2:
2. Relation 1: R1(E,F), FDs: {E} -> {F}, candidate key: {E}.
3. Relation 2: R2(A,B,C,D,E), remaining FD:s {A,B} -> {C,D,E}, {D} -> {B}.
4. Decompose R2 using FD3:
5. Relation 3: R3(D,B), FDs: {D} -> {B}, candidate key: {D}.
6. Relation 4: R4(A,B,C,D,E), FDs: {A,B} -> {C,D,E}.
7. Final BCNF relations are: R1(E,F), R3(D,B) and R4(A,B,C,D,E).

Complementary work 3c):
1: Check for BCNF violations: FD2: {E} -> {F} violates BCNF because {E} is not a superkey. FD3: {D} -> {B} violates BCNF because {D} is not a superkey.
2: Decompose using FD2: Create relation 1: R1(E, F) with FD: {E} -> {F}. Candidate key: {E}. Our remaining relation: R2(A, B, C, D, E) with FDs: {A, B} -> {C, D, E}, {D} -> {B}.
3: Decompose R2 using FD3: Create relation 2: R3(D, B) with FD: {D} -> {B}. Candidate key: {D}. Remaining relation: R4(A, C, D, E) with FD: {A, D} -> {C, E}.
So then our final BCNF relations are: R1(E, F) with FD: {E} -> {F}, candidate key: {E}. R3(D, B) with FD: {D} -> {B}, candidate key: {D}. R4(A, C, D, E) with FD: {A, D} -> {C, E}, candidate key: {A, D}.

Task 4

Consider the relation schema R(A, B, C, D, E) with the following FDs

FD1: {A,B,C} → {D,E}

FD2: {B,C,D} → {A,E}

FD3: {C} → {D}

a) Show that R is not in BCNF.

b) Decompose R into a set of BCNF relations (describe the process step by step).

a)
1. FD1: {A,B,C} -> {D,E}. Determinant {A,B,C} is a superkey, because no violation.
2. FD2: {B,C,D} -> {A,E}. Determinant (B,C,D} is not a superkey, because it violates BCNF.

Complementary work 4a):
To determine if R is in BCNF, we look at each FD to check if determinant is a superkey:
FD1: {A, B, C} -> {D, E}: And the determinant {A, B, C} can determine all attributes in R, so it is a superkey. No BCNF violation. FD2: {B, C, D} -> {A, E}: Compute {B, C, D}^+: Start with {B, C, D}. Using FD2: {B, C, D} -> {A, E}, add {A, E}. Using FD3: {C} -> {D}, {B, C, D} already includes {D}. {B, C, D}^+ = {A, B, C, D, E} = R. Since {B, C, D} is a superkey, there is no violation for FD2. FD3: {C} -> {D}: Compute {C}^+: We start with {C}. Using FD3: {C} -> {D}, add {D}. {C}^+ = {C, D}. Since {C}^+ does not include all attributes of R, {C} is not a superkey. FD3 violates BCNF. So we draw the conclusion that R is not in BCNF because FD3: {C} -> {D} violates the BCNF condition.

b)

1. Decompose using FD2:
2. Relation 1: R1(B,C,D,A,E), FDs: {B,C,D} -> {A,E}, candidate key: {B,C,D}.
3. Relation 2: R2(A,B,C), remaining FDs: {A,B,C} -> {D,E}.
4. Decompose R2 using FD1:
5. Relation 3: R3(A,B,C,D,E), FDs: {A,B,C} -> {D,E}.
6. Final BCNF relations are: R1(B,C,D,A,E) and R3(A,B,C,D,E).

Complementary work 4b):
Step 1: Check for BCNF violations: FD3: {C} -> {D} violates BCNF because {C} is not a superkey.
Step 2: Decompose using FD3: We create relation 1: R1(C, D) with FD: {C} -> {D}. Candidate key: {C}. Remaining relation: R2(A, B, C, E) with FDs: FD1 which is restricted to R2: {A, B, C} -> {E}. FD2 which is restricted to R2: {B, C} -> {A, E}.
Step 3: We check R2 for BCNF violations: FD2 restricted: {B, C} -> {A, E} violates BCNF because {B, C} is not a superkey.
Step 4: We decompose R2 using FD2: so we create relation 2: R2(B, C, A, E) with FD: {B, C} -> {A, E}. Candidate key: {B, C}. Our remaining relation: R3(A, B, C) with FD: {A, B, C} -> {E}. Then we have our final BCNF relations:
R1(C, D) with FD: {C} -> {D}, candidate key: {C}.
R2(B, C, A, E) with FD: {B, C} -> {A, E}, candidate key: {B, C}.
R3(A, B, C) with FD: {A, B, C} -> {E}, candidate key: {A, B, C}.